

Idealized Node Configuration

ASL3 + DVSwitch + USRP2M17 bridge node

Reference for a clean new build (Pi Zero 2 W and full build)

ASL-DVS · prepared with 44helper 0.0.166 · 2026-10-07

No callsigns, IDs, node numbers, passwords, coordinates or addresses appear in this document. Every identity value is a placeholder in <angle brackets>.

1. Purpose and how to read this

This is the target configuration a freshly built ASL-DVS bridge node should end up with: stock ASL3 and DVSwitch files, changed only where the bridge job needs it. It comes from auditing a working Pi Zero 2 W node against the upstream stock files. Where the working node differed from stock, this document says which side it recommends and why.

Identity stays out. Callsigns, DMR/NXDN/other IDs, node numbers, passwords, frequencies and coordinates are entered with `asl-menu` and `dvs`, never in presets. Placeholders used below:

Placeholder	Meaning
<CALL>	your callsign
<NODE>	your AllStarLink node number (real node)
<DMRID> <RPTID> <NXDNID>	your DMR, repeater and NXDN IDs
<PASS>	any password or secret
<LAT> <LON> <LOCATION>	station position and place name

Everything the Config tab in 44helper can check or apply (Standard configuration, Gateways, Port check, What is listening now) is marked `44helper`. Items marked `manual` are not changed by 44helper.

2. Design principles

1. **Stock first.** Keep every stock value unless the bridge job needs a change. Fewer differences means easier support and easier upgrades.
2. **Bridge nodes do not key a radio.** The private USRP nodes run with no hang time and no courtesy tones, so audio passes straight through with no added delay.
3. **Everything between programs stays on 127.0.0.1.** Only IAX, SSH and the web pages are reachable from the network.
4. **One port, one owner.** No two programs may listen on the same UDP port (see section 4).
5. **Services start at boot.** Every gateway the build installs is enabled, so nothing needs starting by hand after a power cycle.
6. **New builds log fully; running nodes log quietly.** Debug is on while building and tuning, and can be turned off later.

3. Audio path

```
AllStarLink network
| IAX2 (UDP 4569)
[ ASL3 / app_rpt ]----- real node <NODE>      (radio hardware, optional)
| USRP (UDP, localhost only)
+-- private node 1999 -- Analog_Bridge --- MMDVM_Bridge --- DMR network
|                                     |
|                                     | DVSswitch.ini      YSFGateway / P25Gateway /
|                                     | (per-mode ports)  NXDNGateway / ircDDBGateway
|
+-- private node 1917 -- USRP2M17 ----- M17 reflector
```

Node numbers 1000-1999 are private and never leave the machine. 1999 is the DVSwitch bridge and 1917 is the M17 bridge; these two numbers are conventions, not secrets, and stay the same on every build.

4. Port map (all UDP unless noted)

Link	Ports	Reach	Notes
USRP to Analog_Bridge	34001 / 32001	127.0.0.1	node 1999 ↔ Analog_Bridge
USRP to USRP2M17	34008 / 32008	127.0.0.1	node 1917 ↔ USRP2M17
Analog_Bridge to MMDVM_Bridge (DMR)	31100 / 31103	127.0.0.1	per mode: DMR 311xx, D-Star 321xx, NXDN 331xx, P25 341xx, YSF 351xx, STFU 361xx, ASL 301xx
MMDVM_Bridge System Fusion	3200 / 4200	127.0.0.1	MMDVM_Bridge ↔ YSFGateway
MMDVM_Bridge P25	32010 / 42020	127.0.0.1	MMDVM_Bridge ↔ P25Gateway
MMDVM_Bridge NXDN	14021 / 14020	127.0.0.1	MMDVM_Bridge ↔ NXDNGateway
MMDVM_Bridge D-Star	20011 / 20010	127.0.0.1	MMDVM_Bridge ↔ ircDDBGateway
MMDVM_Bridge DMR local	62032	outbound	local source port toward the DMR master
USRP2M17	17010 (local) / 17000 (remote)	127.0.0.1 target	M17 Network section; 17010 avoids the 32010 P25 clash
Command ports	6073 YSF, 6074 P25, 6075 NXDN, 54321 D-Star	localhost only	remote-control ports: never open in the firewall
IAX2	4569	network	inbound AllStarLink connections
Asterisk manager	5038 (TCP)	127.0.0.1	local only

Pairing rule. Each side's *listen* port must equal the other side's *send* port. 44helper's Port check card reads every file and flags any pair that does not match, and any port two programs both try to own.

Known clash fixed. M17 used to listen on 32010, the same port MMDVM_Bridge uses for P25. USRP2M17 now uses 17010.

5. ASL3 files (/etc/asterisk)

5.1 rpt.conf

Keep the stock file. Only the node sections are edited. The dashboard Phone tab adds its own marked block; do not edit between its `>>>` and `<<<` markers.

```
[nodes]
<NODE> = radio@127.0.0.1:4569/<NODE>,NONE
1999   = radio@127.0.0.1:4569/1999,NONE
1917   = radio@127.0.0.1:4569/1917,NONE

[events-<NODE>](events-main,events-hotspotradio-usb)

[<NODE>](node-main)
rxchannel = SimpleUSB/<NODE>          ; or Pseudo / dsp device for a hotspot
events    = events-<NODE>
statpost_url = http://stats.allstarlink.org/uhandler
idrecording = |i<CALL>
; stock node-main values apply: duplex 2, hangtime 2000,
; althangtime 4000, telemdefault 2

[1999](node-main)                      ; DVSwitch bridge
rxchannel  = USRP/127.0.0.1:34001:32001
idrecording = |i<CALL>
duplex     = 0
hangtime   = 0
althangtime = 0
holdofftelem = 1
telemdefault = 0
telemdynamic = 0
linktolink  = no
nounkeyct   = 1
tx_timeout  = 170000

[1917](node-main)                      ; M17 bridge
rxchannel  = USRP/127.0.0.1:34008:32008
idrecording = |i<CALL>
duplex     = 0
hangtime   = 0
althangtime = 0
holdofftelem = 1
telemdefault = 0
telemdynamic = 0
linktolink  = no
nounkeyct   = 1
tx_timeout  = 170000
```

Why. A bridge has no radio to protect, so hang time, tail messages and courtesy tones only add delay. `linktolink = no` must carry a value; a bare `linktolink =` is ambiguous. [44helper](#) Standard configuration lists every difference from these values with Apply and Undo.

IAX port. Stock is 4569. Some nodes were built with 4579 and still work, but the firewall then needs 4579 instead of the stock IAX service. Recommended: use 4569 everywhere. If you keep 4579, change the `[nodes]` lines and `iax.conf` together and open that port. [manual](#)

5.2 iax.conf

```
[general]
bindport = 4569
disallow = all
allow = ulaw
allow = adpcm
allow = gsm
jitterbuffer = yes
forcejitterbuffer = yes
dropcount = 2
maxjitterbuffer = 1000
maxjitterinterps = 10
resynchthreshold = 500
maxexcessbuffer = 80
minexcessbuffer = 10
jittershrinkrate = 1
tos = 0x1E
autokill = yes
delayreject = yes
maxcallnumbers_nonvalidated = 10

[radio]
type = user
disallow = all
allow = ulaw
allow = adpcm
allow = gsm
codecpriority = host
context = radio-secure
transfer = no
```

The `[iaxrpt]` user in the stock file is only for the IAXRpt client. Remove it if you do not use that client (section 9).

5.3 modules.conf

Stock autoload-off list. The bridge needs these loaded, and 44helper checks for them:

```
[modules]
autoload = no
require = app_rpt.so
require = chan_iax2.so
require = bridge_softmix.so
require = chan_bridge_media.so
load = chan_simpleusb.so           ; only if a USB sound-card radio is used
load = chan_usrp.so
noload = chan_usbradio.so         ; unused and conflicts with simpleusb
noload = chan_echolink.so         ; unless EchoLink is configured
noload = chan_dahdi.so
noload = chan_voter.so
noload = chan_tlb.so
noload = chan_mobile.so
noload = chan_ooh323.so
; ... the rest of the stock load/noload list is unchanged
```

`require` stops Asterisk starting if the module is missing, which is what you want for `app_rpt`, `chan_iax2` and the two bridge modules. Unloading `chan_usbradio` is an optional efficiency item.

[44helper](#)

5.4 extensions.conf, manager.conf, echolink.conf, rpt_http_registrations.conf

Keep the stock files. The only addition is the Phone-tab block in `extensions.conf`, which the dashboard manages. `manager.conf` must bind to 127.0.0.1 only:

```
[general]
enabled = yes
port = 5038
bindaddr = 127.0.0.1

[<CALL>]
secret = <PASS>
read   = all,system,call,log,verbose,command,agent,user,config
write  = all,system,call,log,verbose,command,agent,user,config
```

`rpt_http_registrations.conf` holds the node registration (node number, password, server list) and is filled by `asl-menu`. Do not preset it.

6. DVSwitch files

6.1 Analog_Bridge.ini (/opt/Analog_Bridge)

```
[GENERAL]
logLevel = 2                ; 1 while building, 2 when quiet
exportMetadata = true
transferRootDir = /tmp
subscriberFile = /var/lib/dvswitch/subscriber_ids.csv
decoderFallBack = true
useEmulator = false
emulatorAddress = 127.0.0.1:2470
pcmPort = 2222

[AMBE_AUDIO]
address = 127.0.0.1
txPort = 31103              ; set by "dvswitch.sh mode <MODE>"
rxPort = 31100              ; set by "dvswitch.sh mode <MODE>"
ambeMode = DMR
minTxTimeMS = 2500
gatewayDmrId = <DMRID>
repeaterID = <RPTID>
txTg = 0
txTs = 2
colorCode = 1

[USRP]
address = 127.0.0.1
txPort = 32001
rxPort = 34001
usrpAudio = AUDIO_USE_GAIN
usrpGain = 2
usrpAGC = -15,8,80
tlvAudio = AUDIO_USE_GAIN
tlvGain = 1.1

[DV3000]
address = /dev/ttyUSB0      ; only with an AMBE3000 stick
baud = 460800
serial = true
```

The [AMBE_AUDIO] ports depend on the current mode: DMR 31103/31100, D-Star 32103/32100, NXDN 33103/33100, P25 34103/34100, YSF 35103/35100, STFU 36103/36100. Do not edit them by hand; use `dvswitch.sh mode`. 44helper's pairing check follows the current mode so it does not raise false warnings. [44helper](#)

6.2 DVSwitch.ini (/opt/MMDVM_Bridge)

Stock. Per-mode ports (txPort/rxPort) as in section 4, plus the remote command ports: D-Star 54321, NXDN 6075, P25 6074, YSF 6073. Leave them as stock and keep them blocked at the firewall.

6.3 MMDVM_Bridge.ini (/opt/MMDVM_Bridge)

```
[General]
Callsign=<CALL>
Id=<DMRID>
Timeout=180
Duplex=0

[Info]
RXFrequency=<FREQ>
TXFrequency=<FREQ>
Power=1
Latitude=<LAT>
Longitude=<LON>
Height=0
Location=<LOCATION>
Description=MMDVM_Bridge

[Log]
DisplayLevel=1
FileLevel=2
FilePath=/var/log/mmdvm
FileRoot=MMDVM_Bridge

[Modem]
Port=/dev/null
Trace=0
Debug=0

[D-Star]
Enable=1
Module=B
[DMR]
Enable=1
ColorCode=1
EmbeddedLCOnly=1
DumpTADData=0
[System Fusion]
Enable=1
[P25]
Enable=1
NAC=293
[NXDN]
Enable=1
RAN=1
Id=<NXDNID>

[D-Star Network]
Enable=1
GatewayAddress=127.0.0.1
GatewayPort=20010
LocalPort=20011
Debug=1 ; 0 once tuned
[DMR Network]
Enable=1
Address=<DMR_MASTER>
Port=62031
Jitter=360
```

```

Local=62032
Password=<PASS>
Slot1=0
Slot2=1
Debug=1 ; 0 once tuned
[System Fusion Network]
Enable=1
LocalAddress=127.0.0.1
LocalPort=3200
GatewayAddress=127.0.0.1
GatewayPort=4200
Debug=1 ; 0 once tuned
[P25 Network]
Enable=1
GatewayAddress=127.0.0.1
GatewayPort=42020
LocalPort=32010
Debug=1 ; 0 once tuned
[NXDN Network]
Enable=1
GatewayAddress=127.0.0.1
GatewayPort=14020
LocalPort=14021
Debug=1 ; 0 once tuned

```

Change from the working node: [System Fusion Network] LocalAddress is 127.0.0.1 here, not 0.0.0.0. Everything else in this program talks only to localhost, so there is no reason to listen on every interface. **Fusion/NXDN/P25 are all enabled**; the modem is a null device because this is a software bridge.

Debug. A new build runs with network Debug=1 and Analog_Bridge logLevel=1 so the first calls can be traced. Once the node is proven, set the network Debug=0 and logLevel=2. This is one optional item in 44helper. [44helper](#)

6.4 Gateways

YSFGateway.ini (/opt/YSFGateway)

```
[General]
Callsign=<CALL>
Suffix=RPT
Id=<ID>
RptAddress=127.0.0.1
RptPort=3200
LocalAddress=127.0.0.1
LocalPort=4200
WiresXMakeUpper=1
WiresXCommandPassthrough=0
Daemon=0
[Log]
DisplayLevel=1
FileLevel=2
FilePath=/var/log/mmdvm
FileRoot=YSFGateway
[Network]
InactivityTimeout=0
Revert=0
Debug=0
[YSF Network]
Enable=1
Port=4200
```

P25Gateway.ini (/opt/P25Gateway)

```
[General]
Callsign=<CALL>
RptAddress=127.0.0.1
RptPort=32010
LocalPort=42020
Daemon=0
[Network]
Port=42010
HostsFile1=/var/lib/mmdvm/P25Hosts.txt
HostsFile2=/var/lib/mmdvm/private_P25Hosts.txt
ReloadTime=60
ParrotAddress=127.0.0.1
ParrotPort=42011
P252DMRAddress=127.0.0.1
P252DMRPort=42012
InactivityTimeout=0
Debug=0
[Remote Commands]
Enable=1
Port=6074
```

NXDNGateway.ini (/opt/NXDNGateway)

```
[General]
Callsign=<CALL>
Suffix=NXDN
RptProtocol=Icom
RptAddress=127.0.0.1
RptPort=14021
LocalPort=14020
Debug=0
Daemon=0
[Network]
Port=14050
HostsFile1=/var/lib/mmdvm/NXDNHosts.txt
HostsFile2=/var/lib/mmdvm/private_NXDNHosts.txt
ReloadTime=60
ParrotAddress=127.0.0.1
ParrotPort=42021
NXDN2DMRAddress=127.0.0.1
NXDN2DMRPort=42022
InactivityTimeout=0
Debug=0
[Remote Commands]
Enable=1
Port=6075
```

Each gateway's `RptPort` must equal MMDVM_Bridge's `LocalPort` for that mode, and its `LocalPort` must equal MMDVM_Bridge's `GatewayPort`. These are the pairs the Port check follows. `Demon=0` is correct because systemd supervises the process. [44helper](#)

ircDDBGateway (/etc/ircddbgateway)

Stock. Gateway linking and the remote-control port stay on the ports the file already has (D-Star MMDVM link 20010/20011, remote 54321). The callsign, gateway password and ircDDB login are identity and are entered through `dvs`.

6.5 USRP2M17.ini

```
[M17 Network]
Callsign=<CALL>
Address=<REFLECTOR_ADDRESS>
Name=<REFLECTOR_NAME>
LocalPort=17010
DstPort=17000
GainAdjustdB=3
Daemon=1
Debug=0

[USRP Network]
Address=127.0.0.1
DstPort=32008
LocalPort=34008
GainAdjustdB=3
Debug=0

[Log]
DisplayLevel=0
FileLevel=0
FilePath=/var/log/usrp
FileRoot=USRP2M17
```

Written by the dashboard M17 page. [manual](#) (the dashboard owns it).

7. Services

Service	State	Note
asterisk	enabled, running	core
allmon3	enabled, running	status page; keep its node list in step with rpt.conf (44helper presets do)
analog_bridge	enabled, running	DVSwitch audio
mmdvm_bridge	enabled, running	DVSwitch modem layer
ysfgateway	enabled, running	Fusion
ircddbgatewayd	enabled, running	D-Star
p25gateway	enabled , running	installed disabled by default; enable on every new build
nxdngateway	enabled , running	installed disabled by default; enable on every new build
stfu	enabled only if BrandMeister STFU is used	this and DMR share the BrandMeister login
usrp2m17 (and M17 gateway services)	enabled, running	started when the dashboard M17 page is entered, restarted on leave

Enable-and-start at boot for the gateways is one button on the Config tab (Gateways card), with Undo.

44helper

8. Firewall (firewalld)

Allow in from the network: **IAX2 on the IAX port (4569)**, **SSH**, and the **web ports** you use (dashboard, allmon3). Everything else is closed.

```
# apply once, then reload
sudo firewall-cmd --permanent --zone=<ZONE> --add-service=ssh
sudo firewall-cmd --permanent --zone=<ZONE> --add-port=4569/udp
sudo firewall-cmd --permanent --zone=<ZONE> --add-port=<DASHBOARD_PORT>/tcp
sudo firewall-cmd --reload
sudo firewall-cmd --zone=<ZONE> --list-all
```

Do not open UDP 6073, 6074, 6075 or 54321 (gateway command ports), TCP 5038 (manager), or any broad range such as 10000-20000/udp. A range like that quietly exposes 17010 (M17) and 14021 (NXDN). If the zone has one, narrow it. **Do not open** echolink service ports unless EchoLink is configured.

Everything between programs binds to 127.0.0.1, so a closed firewall does not affect the audio path. The Config tab's *What is listening now* card shows each listening socket, whether the firewall lets it in, and which program owns it. 44helper

9. Optional and removable items

- **iaxrpt user** in iax.conf and the [iaxrpt] context in extensions.conf: remove if the IAXRpt client is not used.
- **chan_usbradio**: unload (section 5.3).
- **EchoLink**: leave the module off and the service ports closed unless it is set up.
- **Debug**: turn off after tuning (section 6.3).
- **Gateways not used**: leave their programs enabled anyway on a bridge; they cost almost nothing idle and the mode switch then just works.

10. Applying this with 44helper

Task	Where
Save the as-installed files	Config tab: Save Restore point (or the Save step at the end of each install tab). Do it once, after every system you want is installed.
Wiring and private nodes	Config tab: Travel Node preset (wiring only, no identity).

Differences from stock	Standard configuration card: review, Apply, Undo.
Gateways at boot	Gateways card: Enable and start at boot.
Ports	Port check and What is listening now cards.
Go back	Restore from the Restore point, a save point, or an automatic copy.

11. Verification checklist

1. `asterisk -rx 'rpt show variables 1999'` shows the node and USRP channel; same for 1917.
2. `sudo ss -H -tulnp`: every program in section 4 is listening, on 127.0.0.1 except IAX, SSH and web.
3. No UDP port is owned by two programs.
4. Config tab Port check: every pair matches, no clashes.
5. Config tab Standard configuration: no unexpected differences.
6. Services in section 7 are enabled, running, and still running after a reboot.
7. From outside the node, ports 6073-6075, 54321 and 5038 are closed.
8. A test call in each enabled mode passes audio in both directions.

12. What this document does not cover

Radio hardware settings (simpleusb.conf levels, squelch, CTCSS), EchoLink credentials, ircDDB and BrandMeister logins, frequencies and coordinates, and the Phone-tab dialplan (see the Phone Dialplan Guide). These are identity or site specific.